D Download and Installation of LTspice and the Power-System Toolboxes

D.1 Purpose and Required Downloads

The examples in this book use LTspice Version 24.0.12 together with two power-system toolboxes:

- **Transient toolbox:** models and examples for electromagnetic transient simulations of high-energy power networks;
- **Steady-state toolbox:** models and examples for load-flow and steady-state simulations.

This book concentrates on transient simulations. The steady-state toolbox is optional and is required only when the corresponding load-flow models are to be used.

Files to download

Download the following archive from the project homepage: The transient and loadflow Toolboxes and the LTspice 24.0.12 simulator installation file in one archive `LTspice.zip`; The archive is large because it also contains the now free MicroCap12 simulator with Library file for loadflow and transient simulation and with some examples! This file seems a little messy because it's the development version. You can also download LTspice 24.0.12 from Github Search there for "loadflow ltspice". If you only need the LTspice toolbox files preferably option is to download the archive from the LTspice usergroup in the files-z_yahoo section `adventures with loadflow`. NOTE: The LTspice simulator 24.0.12 must then be downloading separately

Project homepage:

```
my homepage: <www.friedrichschmidt.at>  or
LTspice usergroup files at: <https://groups.io/g/LTspice/files/z_yahoo/
    adventures%20with%20loadflow>  or on GitHub ( search for "loadflow ltspice" )
```

Important D.1 – (Version used by this book)

The models and screenshots described in this book were prepared for LTspice Version 24.0.12. Newer releases may have problems displaying simulation results in the waveform viewer. Retain the 24.0.12 installer if exact reproducibility is required.

D.2 Installing LTspice 24.0.12

1. Extract the downloaded LTspice archive to a temporary folder.
2. Right-click the installer and select Run as administrator.
3. Complete the installation using the default installation options.
4. Start LTspice once and close it again before copying the toolboxes. This confirms that the simulator has been installed correctly and allows it to create its normal user configuration files.

Administrator rights. Administrator rights are required only for the LTspice installation when the installer writes to protected Windows folders. The toolbox files themselves are installed in the user's Documents folder and should normally not require administrator rights.

D.3 Installing the Toolboxes

In the paths below, replace `<username>` with the actual Windows user name. Extract the transient toolbox into

```
C:\Users\<username>\Documents\LTspice
```

After extraction, the transient-toolbox folder must be

```
C:\Users\<username>\Documents\LTspice\loadflow_transient
```

If the steady-state toolbox is also required, extract it so that the following folder exists:

```
C:\Users\<username>\Documents\LTspice\loadflow_steady_state
```

Important D.2 – (Check the extracted folder level)

Avoid an additional, duplicated directory level such as `LTspice\LTspice\loadflow_transient`. The search paths configured in LTspice must point directly to the folders that contain the toolbox symbols and library files.

Updating an existing installation. If a toolbox folder already exists, make a backup before extracting a newer archive. This preserves locally modified symbols, models, plot settings, and example circuits.

D.4 Configuring Search Paths

Start LTspice and open the configuration dialog with

Tools → Control Panel.

Open the Sym. & Lib. Search Paths tab. Add the transient-toolbox path to both the **Symbol Search Path** and the **Library Search Path**:

```
C:\Users\<username>\Documents\LTspice\loadflow_transient
```

Thus, the same path is entered twice: once in each of the two search-path lists.
If the steady-state toolbox is installed, also add

```
C:\Users\<username>\Documents\LTspice\loadflow_steady_state
```

to both search-path lists.

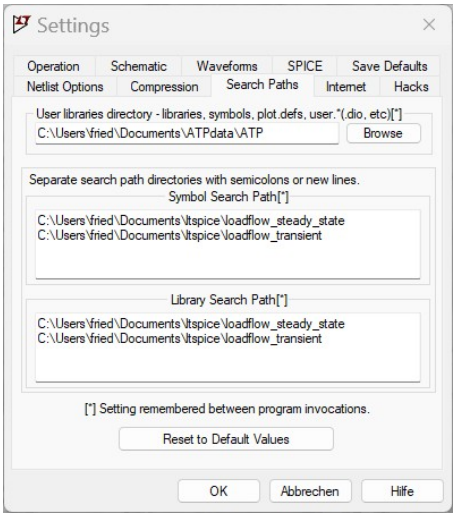


Figure D.1: Symbol and Library search path tab (best for user libraries: point to an empty folder in documents)

Table D.1: Required LTspice symbol and library search paths.

Toolbox	Folder	Add to both lists
Transient	`C:\Users\<username>\Documents\LTspice\loadflow_transient`	Yes
Steady state	`C:\Users\<username>\Documents\LTspice\loadflow_steady_state`	Yes

D.5 Configuring the Simulator

Keep the Control Panel open and configure the following tabs.

D.5.1 SPICE tab

Set the numerical solver to

Solver: Alternate.

The alternate solver is recommended for the large networks, nonlinear magnetic models, switches, and behavioral equations used in the transient toolbox.

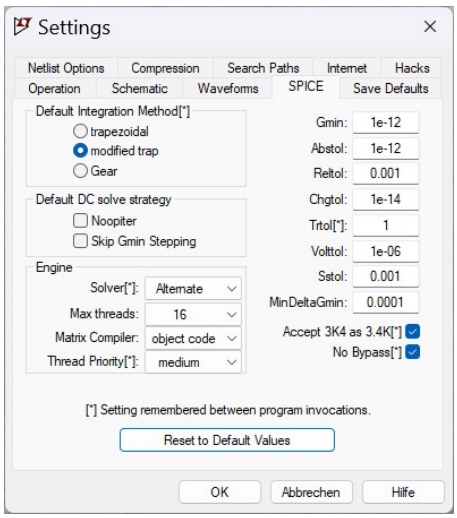


Figure D.2: Spice tab

D.5.2 Save Defaults tab

Enable the first three checkboxes on this tab. The exact labels can vary slightly between LTspice builds. These options ensure that the internal data required by many toolbox plots and hierarchical models are retained.

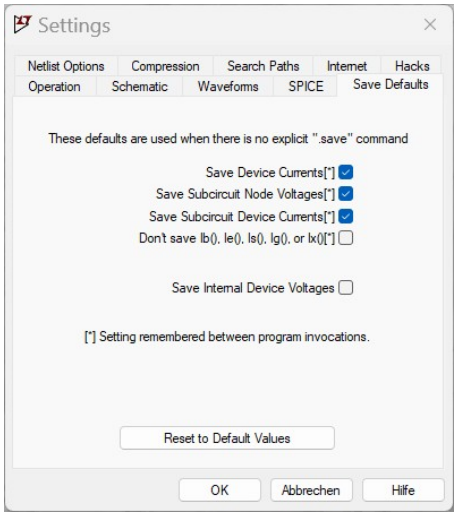


Figure D.3: Save defaults tab

> **Internal subcircuit quantities.** Several examples inspect internal transformer fluxes, winding currents, or hierarchical node voltages. If these traces are unavailable, return to SAVE DEFAULTS and verify that the relevant subcircuit node voltages and currents are saved.

D.5.3 Schematic tab

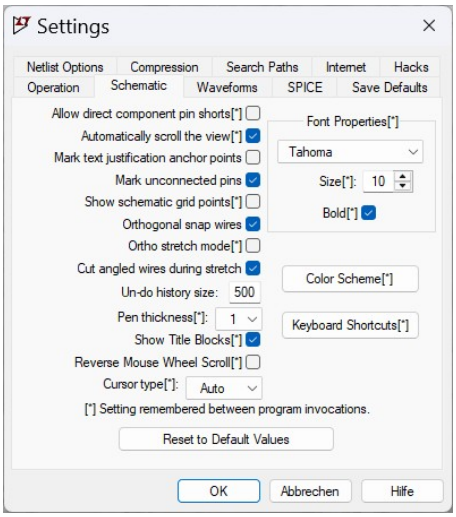


Figure D.4: Schematic tab

Set the schematic-editor font to

Arial, 10 pt, Bold.

D.5.4 Waveforms tab

Set the waveform-viewer font to

Arial, 12 pt, Data trace width=2, Cursor width=2.

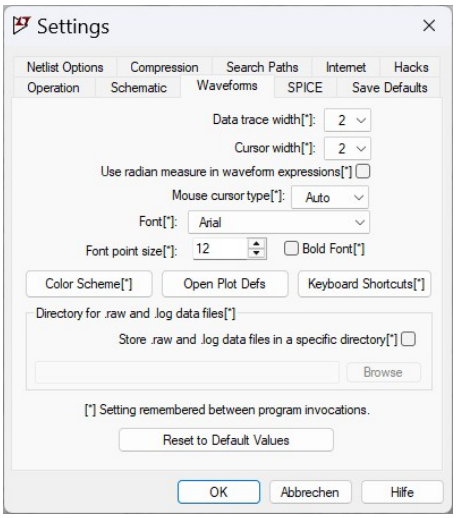


Figure D.5: Waveforms tab

D.5.5 Operation tab

Enable the following options:

- **Marching Waveforms**;
- **Generate Expanded Listing**;
- **Save all open files on start of simulation**;
- **Automatically Delete .raw Files**.

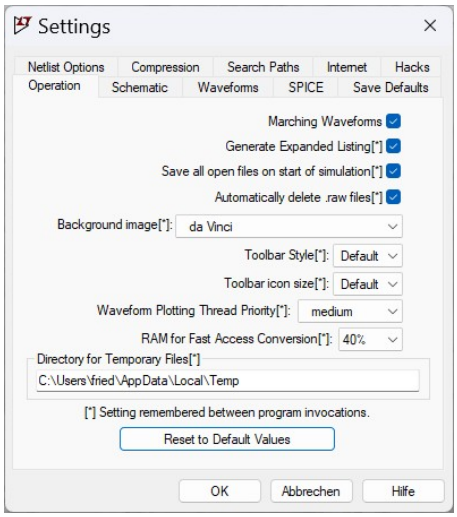


Figure D.6: Operation tab

Remark D.1 – (Automatically deleting RAW files)

RAW files from long transient simulations can become very large. Automatic deletion saves disk space after the waveform window is closed. Disable this option temporarily if a RAW file must be retained for later evaluation or shared with another user.

Table D.2: Summary of the recommended LTspice settings.

Control Panel tab	Recommended setting
Sym. & Lib. Search Paths	Add each installed toolbox folder to both the symbol and library search-path lists.
SPICE	Solver: **Alternate**.
Save Defaults	Enable the first three checkboxes.
Drafting Options	Arial, 10 pt, Bold.
Waveforms	Arial, 10 pt, Bold.
Operation	Enable Marching Waveforms, Generate Expanded Listing, and Automatically Delete .raw Files.

D.6 Verifying the Installation

After closing the Control Panel, restart LTspice and perform the following checks:

1. Open the component-selection dialog and verify that toolbox symbols can be selected.
2. Open one transient-toolbox example circuit.
3. Confirm that LTspice reports no missing symbol or library-file errors.
4. Run a short transient analysis.
5. Verify that the waveform viewer opens and that hierarchical quantities required by the example can be selected.
6. If the steady-state toolbox is installed, open and run one of its load-flow examples as a separate check.

Important D.3 – (If models or symbols are not found)

Check the spelling of the Windows user name, confirm the extracted folder level, and verify that the toolbox path was entered in both search-path lists. After changing a search path, restart LTspice before testing the installation again.

Appendix Summary

Install LTspice 24.0.12 with its default options, extract the transient toolbox to the user's Documents folder, and optionally install the steady-state load-flow toolbox. Add every installed toolbox folder to both the symbol and library search paths. Use the alternate solver, save the internal model data required by the examples, configure consistent schematic and waveform fonts, and enable the recommended operating options. A short example simulation should be run after installation to verify that all symbols and library models are available.